

Validation of novel *CNBP* repeat features in Myotonic Dystrophy Type 2 through comparative long-read sequencing with Oxford Nanopore and PacBio

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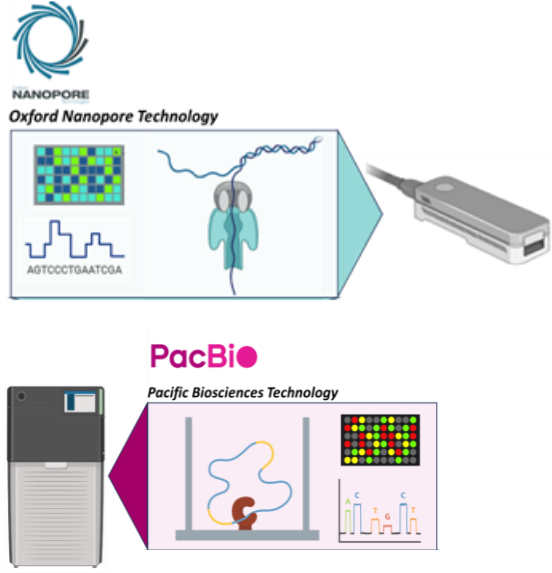


Introduction

Myotonic Dystrophy Type 2 (DM2) is caused by massive and highly mosaic CCTG repeat expansions in the *CNBP* gene. While targeted long-read sequencing has enabled the sequencing of *CNBP* repeats and revealed a novel TCTG interruption pattern (Alfano M. et al, eLife 2022), the computational characterization of this region remains challenging. Currently, there is a lack of clear consensus and benchmarking of bioinformatic pipelines to accurately identify the architecture of these expansions. In this work, our goal was to establish a standardized computational framework through systematic algorithmic benchmarking and cross-platform validation, in order to accurately resolve these expansions

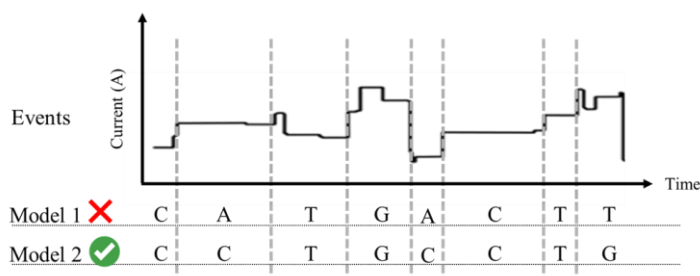
Methods

1. Targeted sequencing using LRS



2. Basecaller benchmarking (ONT)

Which model and tool to use?



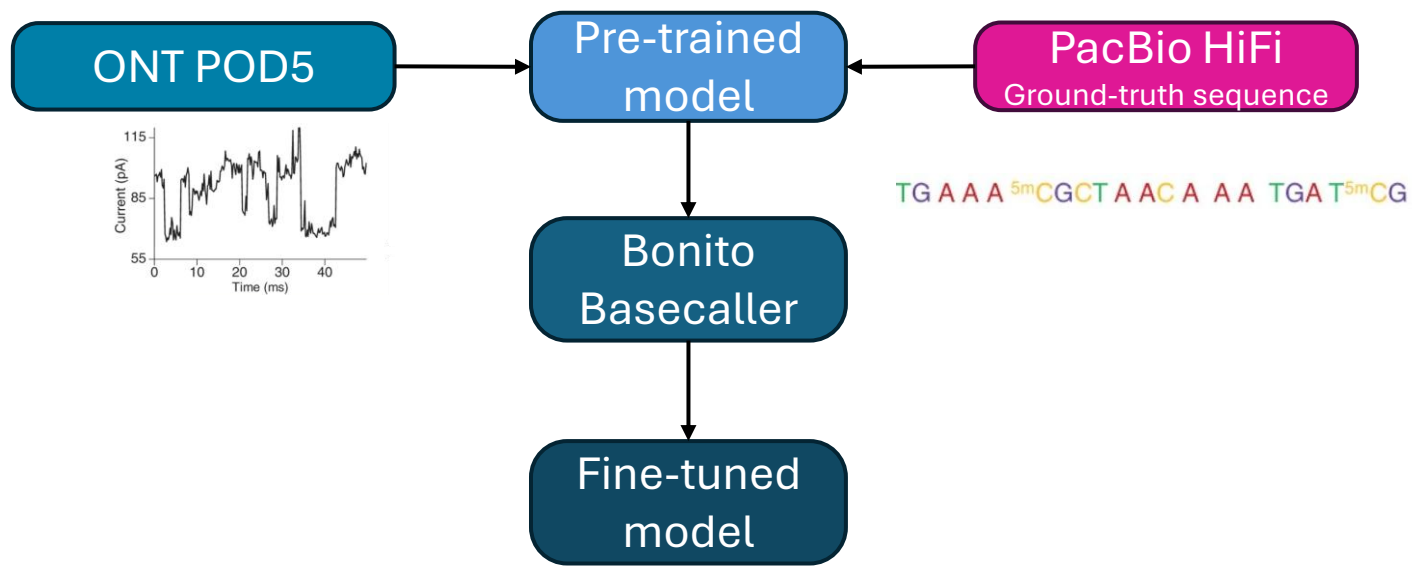
- Dorado basecaller (sup@5, sup@5.2, sup@4.3)
- Bonito basecaller
- Bonito basecaller retrained with PacBio data

3. Downstream analysis

Visualization, repeat size

- Mosaicism visualization
- Maximum/median/minimum sizes alleles
- Motif quantification

4. Training Bonito basecaller to improve the accuracy



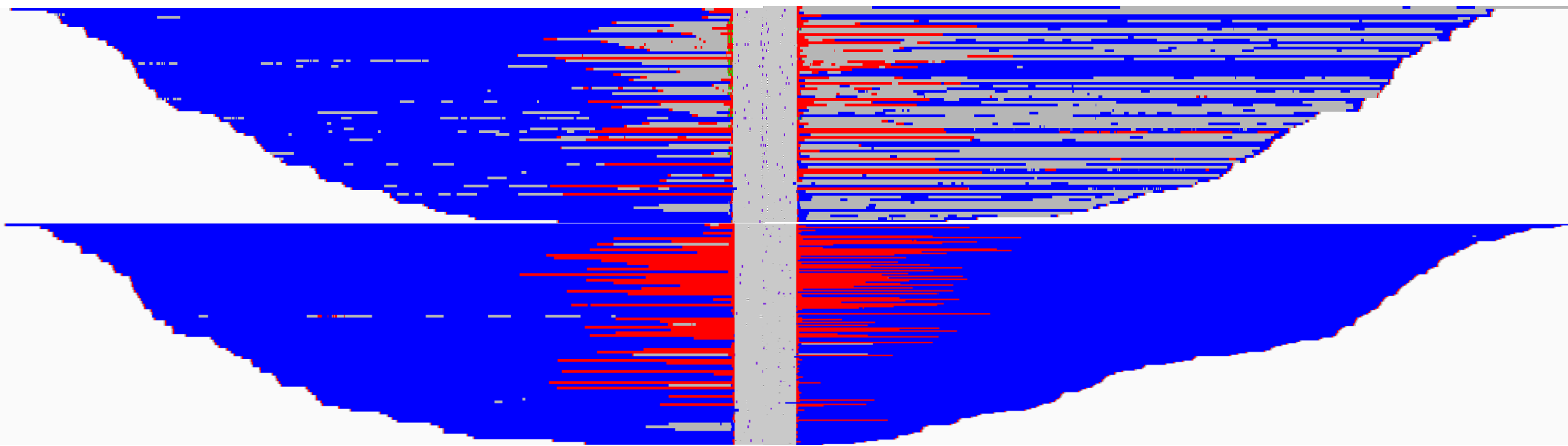
We sequenced the largest DM2 cohort to date, using a Cas9-mediated approach combined with Oxford Nanopore (ONT) and PacBio platforms. To assess the impact of signal decoding on repeat resolution, raw POD5 files from ONT data were basecalled using multiple versions of the Dorado super-accurate model. Complete reads spanning the entire expansion were extracted for downstream benchmarking. We compared two distinct tools for expansion sizing, motif identification, and mosaicism visualization: TRGT and MosaicViewer, developed for repeat expansion analysis with PacBio and ONT, respectively.

In order to improve the accuracy of the models when basecalling highly repetitive motifs, we fine-tuned a pre-trained model using Bonito Basecaller, using the ONT POD5 data and, as their ground-truth reference, the HiFi sequences obtained with PacBio

Results

ONT
Dorado
4.3.0

Bonito
fine-
tuned
model



ONT
Dorado
5.0.0

PacBio
HiFi

Our benchmarking revealed that the choice of the basecalling algorithm version significantly impacts the accuracy of the repeat sequence identification based on ONT data. In particular, specific Dorado versions showed higher concordance with PacBio HiFi data, proving that basecaller version has a great impact on the resolution of repeat motif

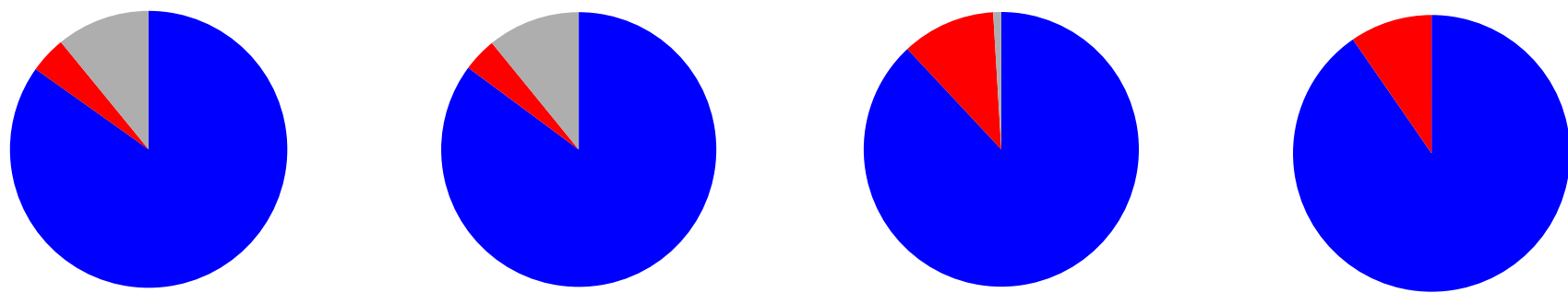
Dorado
5.0.0

Dorado
4.3.0

Bonito
pre-trained

Bonito
fine-tuned

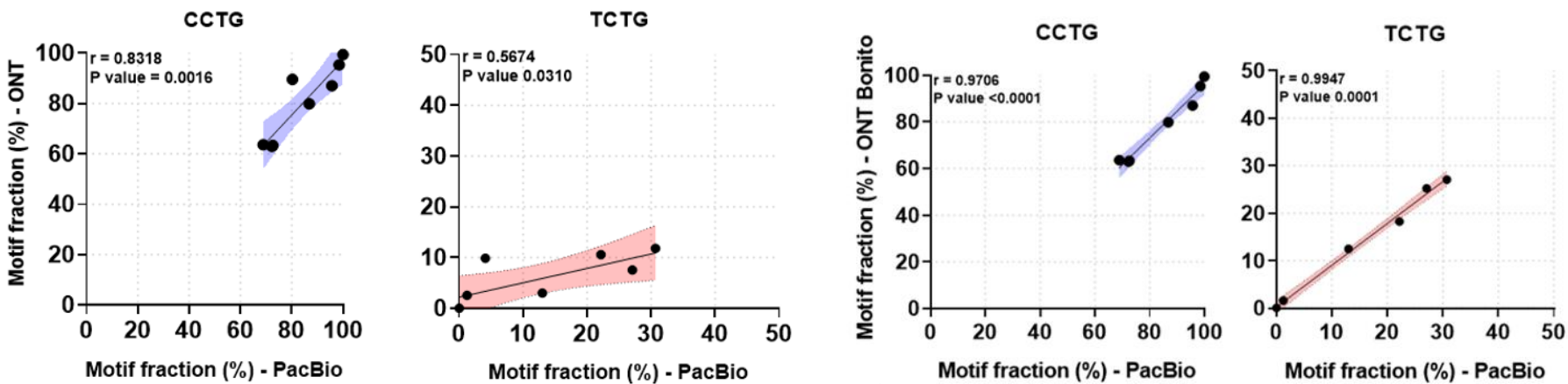
PacBio
HiFi



The proportion of correctly identified motifs progressively increases. Selecting the appropriate Dorado basecaller model version significantly improves repeat motif resolution, and fine-tuning Bonito using PacBio HiFi data as ground-truth further reduces motif misclassification, bringing ONT results closer to PacBio accuracy

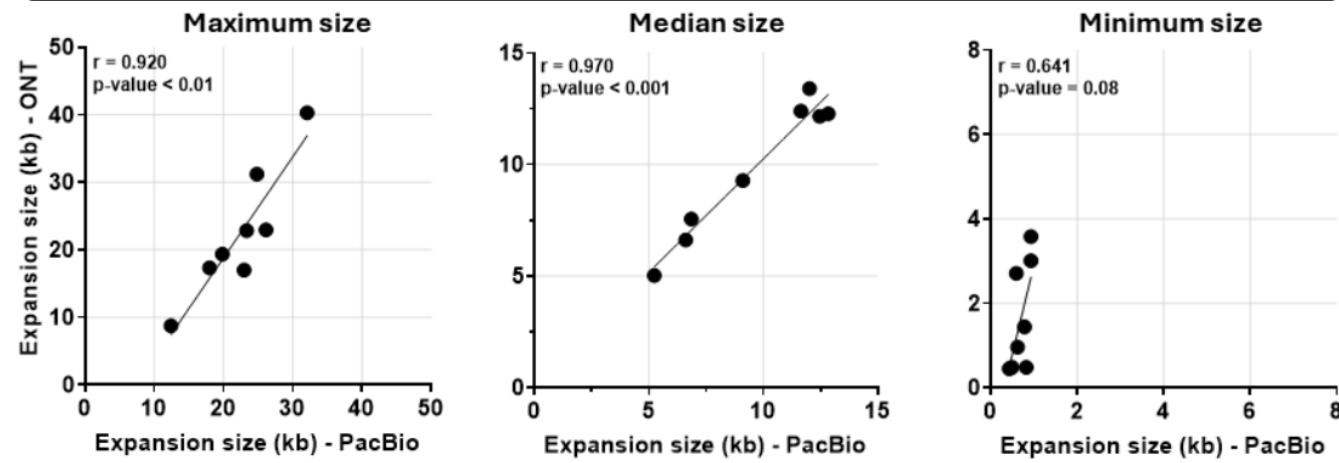
ONT - Dorado vs PacBio

ONT - Bonito fine-tuned vs PacBio

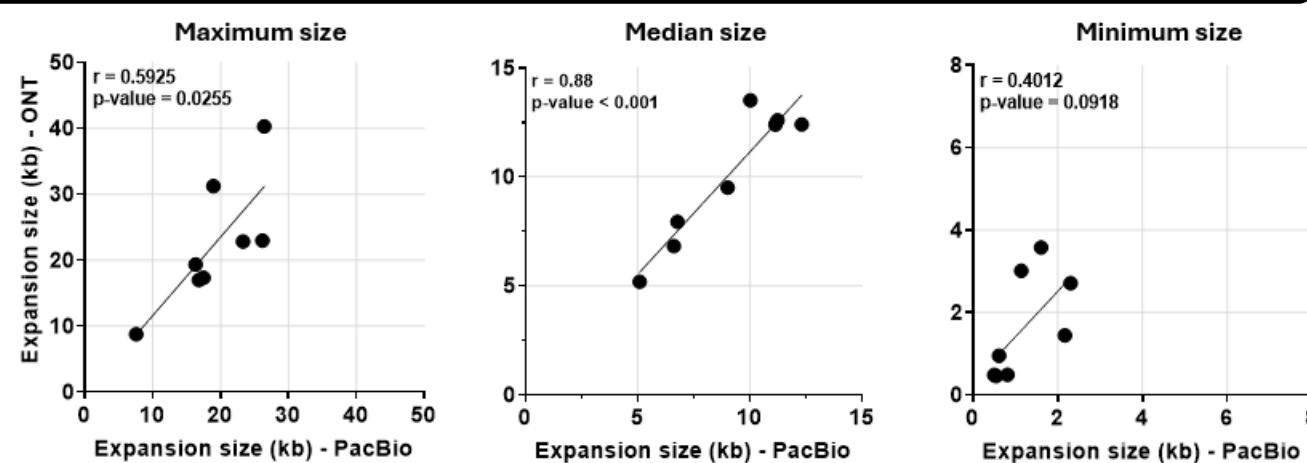


Correlations between the fraction of repeat units across the total number of repeats show an improvement after fine-tuning Bonito

Size correlation without downsampling



Size correlation after downsampling



Maximum and median expansion sizes are comparable across ONT - Dorado and PacBio, although ONT tends to overestimate the minimum expansion size. Downsampling PacBio reads to ONT coverage improves the minimum allele size correlation, while maximum and median keeps unchanged

Conclusions

1. The basecaller model selection critically impacts on the accuracy of the results
2. Dorado is highly comparable with PacBio for every aspect, turning it into an interesting and comparable alternative
3. Novel TCTG motif was confirmed using PacBio data, although ONT's Dorado basecaller shows some difficulties at identifying it
4. Bonito fine-tuned model improved the novel TCTG motif detection, enhancing the resolution achieved with ONT over repetitive motifs